

OptiComp2

Verification register and test report

Every automated test of the instrument-control software, with coverage figures

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August 2026 · 149 tests · 9 modules · 31 test classes

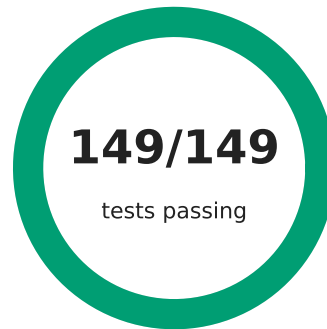
This report lists every automated test in OptiComp2, grouped by module and by what it verifies, together with the coverage figures. The whole suite runs without any hardware — against a fake serial port and a fake spectrometer DLL — so it can be run on a workstation and on the instrument PC alike. It is the executable specification of the safety and correctness properties described in the change list.

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1 Summary

Result



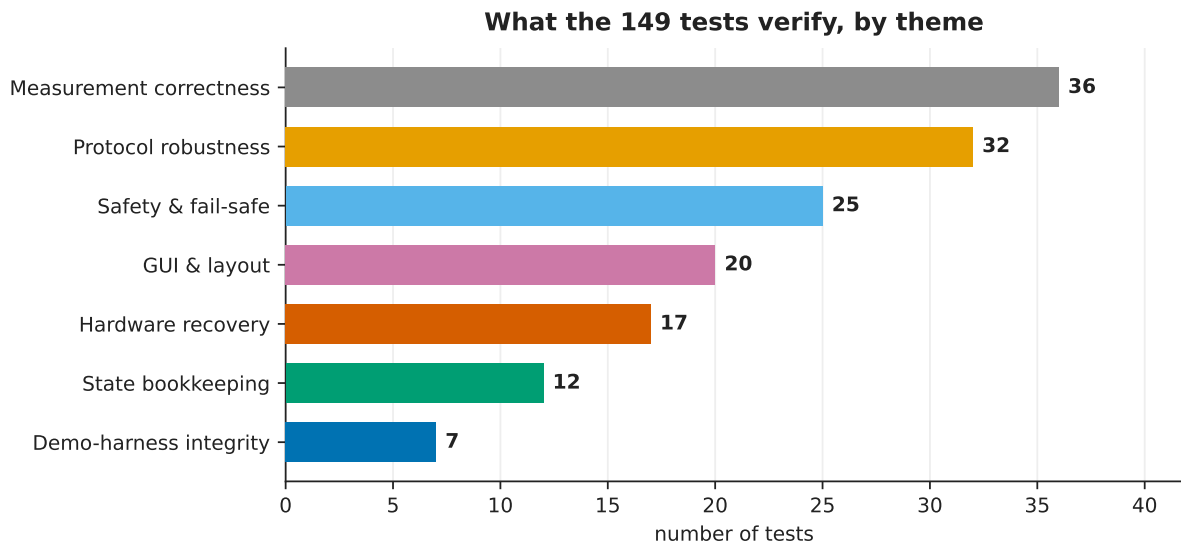
no hardware required - fake serial port + fake DLL - passes on the workstation and the instrument PC (Python 3.9)

The suite comprises **149 tests** in 31 classes across 9 modules, and all pass. Run it with:

```
py -m unittest discover -s tests
```

Every test uses in-memory fakes (a scripted serial port, a fake BWTEKUSB.dll, a demo bus and spectrometer), so no instrument, lamp or motion is involved and the suite is deterministic and safe to run anywhere. Fault injection (swallowed commands, truncated replies, a hung DLL, a corrupt manifest, an operator-absent pause) is used to exercise the recovery and abort paths that matter most on the real instrument.

2 Coverage by theme

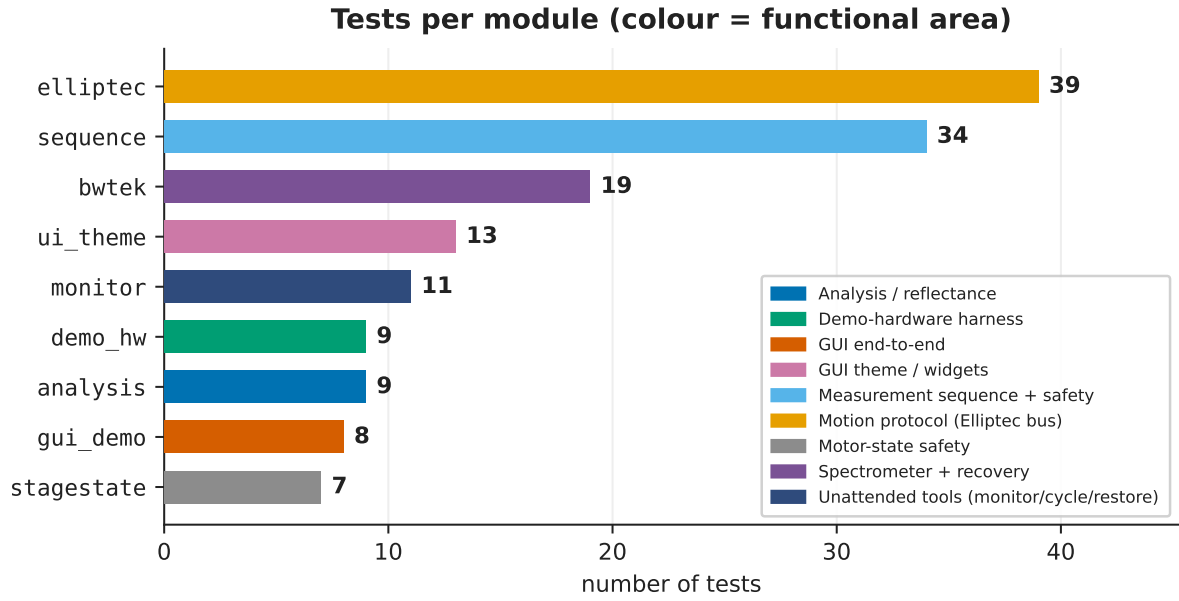


The tests are grouped below by the property they establish:

- **Measurement correctness (36)** — the reflectance relation, dark matching, saturation and invalid-wavelength masking, the Fresnel/BK7 standards, the scan gatekeeper, manifest bookkeeping and integration-time calibration.
- **Protocol robustness (32)** — reply decoding, swallowed-command resend, stray/truncated replies, timeouts, corrective moves and the mechanical-timeout policy on the Elliptec bus.

- **Safety & fail-safe (25)** — the shutter is closed on any error or abort, the fibre arm is protected from homing, soft limits hold, auto integration time aborts on no light, and the state gate blocks motion after an anomaly.
- **GUI & layout (20)** — the theme and widgets, the end-to-end demo tour, the feature- parity reference and the layout audit at two window sizes.
- **Hardware recovery (17)** — the spectrometer reopen / USB-restart ladder and the monitor/cycle tools recovering from an injected read failure.
- **State bookkeeping (12)** — the motor-state record: round-trip, wrap-around comparison, settling tolerance and corrupt-record handling.
- **Demo-harness integrity (7)** — the fakes never touch real hardware and reproduce realistic, seeded spectra.

3 Coverage by module



Module	Functional area	Tests
test_elliptec.py	Motion protocol (Elliptec bus)	39
test_sequence.py	Measurement sequence and safety	34
test_bwtek.py	Spectrometer and recovery	19
test_ui_theme.py	GUI theme and widgets	13
test_monitor.py	Unattended tools (monitor / cycle / restore)	11
test_analysis.py	Analysis and reflectance	9
test_demo_hw.py	Demo-hardware harness	9
test_gui_demo.py	GUI end-to-end (demo hardware)	8
test_stagestate.py	Motor-state safety	7
Total		149

4 Methodology

- **No hardware.** A scripted serial port answers Elliptec queries from a table; a fake DLL returns canned frames and error codes; a demo bus and spectrometer drive the GUI. The production code path is exercised unchanged — only the lowest-level back-end is substituted.
- **Fault injection.** Tests force the failure modes seen on the instrument: a swallowed `ma`, a reply truncated by the read timeout, a stray reply from another module, a mechanical timeout (GS02), a

hung DLL (read `-99`), a corrupt manifest and an operator-absent pause — and assert the resend, discard, abort or recovery that must follow.

- **Safety assertions.** The shutter-closing, protected-home and soft-limit properties are asserted directly (for example, an error raised with the shutter open must leave it closed).
- **GUI.** The demo tour drives the whole workflow in the Tk loop and a layout audit checks that no widget is clipped at 1280×820 or 1100×720 .
- **Both platforms.** The suite passes on a workstation and on the instrument PC (Python 3.9, `py -m unittest discover -s tests`), with no third-party test runner.

5 Full test register

Each test is listed with its one-line purpose (from the test's own comment, or its name where it is self-describing). Test names are the exact method names, for traceability.

Motion protocol (Elliptec bus)

test_elliptec.py · 39 tests

CodecTests

test_hex32_roundtrip	Hex32 roundtrip
test_info_decode_full_serial	Info decode full serial
test_gs_po_decode	Gs po decode

BusTests

test_query_timeout_raises	Query timeout raises
test_info_and_position	Info and position
test_move_waits_while_busy_then_reads_position	Move waits while busy then reads position
test_swallowed_command_is_resent	Real-bus case 2026-08-25: no reply, no motion, GS00 -> resend; second attempt works
test_swallowed_command_gives_up_after_attempts	Swallowed command gives up after attempts
test_wrong_final_position_raises	Wrong final position raises
test_small_residual_triggers_corrective_move	Real-bus case: ELL18 stopped 42 pulses short after a long move; a second 'ma' fixes it
test_persistent_small_residual_is_accepted_with_warning	Persistent small residual is accepted with warning
test_relative_move_correction_uses_absolute_target	Relative move correction uses absolute target
test_lost_reply_but_move_completed	Lost reply but move completed
test_motion_uses_long_timeout_and_restores	Motion uses long timeout and restores
test_deferred_po_arrives_during_gs_poll	Real-bus case: ELL18 silent while moving, completion PO lands in the 'gs' read window
test_truncated_late_po_fragment_is_discarded	Real-bus case 2026-08-25 21:53: the completion PO of module 2 arrived while the input buffer
test_stray_reply_from_other_module_is_discarded	Stray reply from other module is discarded
test_only_stray_replies_raises	Only stray replies raises
test_mechanical_timeout_retried_once	Mechanical timeout retried once
test_sensor_error_at_target_is_accepted	Real-bus case: GS0A after a long move, but gp shows the stage at the target
test_sensor_error_off_target_raises	Sensor error off target raises
test_mechanical_timeout_twice_raises	Mechanical timeout twice raises
test_mechanical_timeout_recovers_within_default_retries	Two GS02s then a good move: the default retry budget (3) rides out a couple of stalls
test_shutter_already_open_is_accepted	Shutter already open is accepted
test_shutter_swallowed_fw_is_resent	Shutter swallowed fw is resent
test_move_error_status_raises	Move error status raises
test_wrong_address_reply_raises	Wrong address reply raises
test_rotation_stage_degrees	Rotation stage degrees

ReplyIntegrityTests *2026-08-26 review: partial lines and out-of-order kinds must never be taken as answers.*

<code>test_truncated_payload_is_an_error</code>	Truncated payload is an error
<code>test_terminated_fragment_is_discarded</code>	Readline() returned a fragment cut by the timeout, the real line follows
<code>test_complete_line_without_terminator_is_not_trusted</code>	Complete line without terminator is not trusted
<code>test_late_po_before_gs_answer_is_skipped</code>	A completion PO arrives just before the answer to our gs
<code>test_home_is_never_repeated_after_mechanical_timeout</code>	Home is never repeated after mechanical timeout
<code>test_relative_move_retries_as_absolute</code>	Relative move retries as absolute

PollRobustnessTests

<code>test_stray_line_burst_during_gs_poll_is_not_fatal</code>	A stray-line burst makes query() raise a generic ElliptecError (not ReplyTimeout);
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ProtectedHomeTests

<code>test_protected_module_refuses_home_unless_forced</code>	Protected module refuses home unless forced
<code>test_raw_home_to_protected_module_is_refused</code>	Raw home to protected module is refused
<code>test_unprotected_module_homes_normally</code>	Unprotected module homes normally
<code>test_negative_position_after_power_cycle_decodes</code>	Negative position after power cycle decodes

Measurement sequence and safety

`test_sequence.py` · 34 tests

BuilderTests

<code>test_gatekeeper</code>	Gatekeeper
<code>test_scan_expansion</code>	Scan expansion
<code>test_reference_calibration_is_80_S</code>	Reference calibration is 80 S

RunnerTests

<code>test_reacquired_tag_replaces_earlier_record</code>	Reacquired tag replaces earlier record
<code>test_full_run_writes_files_and_manifest</code>	DB spectra at the fixed DB integration time, session integration time restored afterwards
<code>test_manifest_appends_across_runs_and_preview_hook</code>	Manifest appends across runs and preview hook
<code>test_wrong_angle_aborts</code>	Wrong angle aborts
<code>test_soft_limit_and_abort</code>	Soft limit and abort

StateBookkeepingTests

<code>test_moves_and_shutter_are_recorded</code>	Moves and shutter are recorded
<code>test_arm_moved_without_us_aborts</code>	Arm moved without us aborts
<code>test_sample_moved_without_us_warns_and_continues</code>	Sample moved without us warns and continues
<code>test_consistent_position_is_silent</code>	Consistent position is silent
<code>test_state_disabled</code>	State disabled

SafetyTests *The Runner must never leave the shutter open or lose the manifest when a run fails.*

<code>test_error_with_shutter_open_closes_it</code>	Error with shutter open closes it
<code>test_abort_at_pause_closes_shutter</code>	Abort at pause closes shutter
<code>test_shutter_known_closed_is_not_touched</code>	Shutter known closed is not touched
<code>test_auto_it_without_light_aborts</code>	Auto it without light aborts
<code>test_darks_per_integration_time_do_not_collide</code>	Darks per integration time do not collide
<code>test_abort_inside_double_beam_restores_integration_time</code>	Abort inside double beam restores integration time
<code>test_pause_without_operator_aborts</code>	Pause without operator aborts
<code>test_soft_limits_helper</code>	Soft limits helper
<code>test_corrupt_manifest_is_kept_aside</code>	Corrupt manifest is kept aside

ReliabilityFixTests *2026-08-27 operability fixes: reconnect the spectrometer and zero every stage at the start*

<code>test_build_reset_is_absolute_and_never_homes</code>	Build reset is absolute and never homes
<code>test_preflight_runs_before_the_first_step</code>	Preflight runs before the first step
<code>test_read_recovers_after_spectrometer_error</code>	Read recovers after spectrometer error
<code>test_read_error_without_recover_hook_propagates</code>	Read error without recover hook propagates

StabilityRegressionTests

<code>test_open_shutter_commits_state_before_move_for_failsafe</code>	If forward() raises mid-open, shutter_open must already read True so that a later
<code>test_close_shutter_clears_state_only_after_completed_close</code>	Close shutter clears state only after completed close
<code>test_load_manifest_tolerates_non_dict_json</code>	A corrupt manifest that parses as a list (not a dict) must not raise AttributeError

HWF02StabilityGateTests

<code>test_reference_calibration_stabilise_flag_controls_wait_step</code>	Reference calibration stabilise flag controls wait step
<code>test_wait_stable_passes_on_a_steady_lamp</code>	In-band, constant output -> drift 0 -> gate clears after 'need' reads (no _auto_it needed)
<code>test_wait_stable_warns_but_never_blocks_on_drift</code>	Wait stable warns but never blocks on drift

AutoITCeilingTests

<code>test_too_dim_target_fails_fast_at_the_hw_ceiling</code>	A non-dark but too-dim target (below the accept band even at the 60 s hardware ceiling) must
<code>test_flat_dark_still_aborts_with_no_light</code>	Flat dark still aborts with no light

Spectrometer and recovery

`test_bwtek.py` · 19 tests

Tests

<code>test_wavelength_polynomial_matches_original</code>	Wavelength polynomial matches original
<code>test_open_set_read_close</code>	Open set read close
<code>test_no_device_raises</code>	No device raises
<code>test_read_error_raises_not_zeros</code>	Read error raises not zeros

test_read_before_open_raises

Read before open raises

ITCalibrationTests

test_scales_linearly_towards_target

1000 ms gave 53030 with baseline 900 -> expect $\sim 1000 * (0.85 * 65535 - 900) / (53030 - 900)$

test_saturated_halves

Saturated halves

test_dark_grows_and_clamps

Dark grows and clamps

StabilityRegressionTests

test_stall_warning_with_no_integration_time_does_not_crash

Integration_ms is None (never set) and the read is "slow": the stall-warning branch

test_usb_instance_ids_are_locale_independent

Non-English Windows translates the "Instance ID:" label; the parser must key on the

PnpSafetyTests

test_only_spectrometer_ids_may_be_restarted

Only spectrometer ids may be restarted

RecoveryTests

test_reopen_reapplies_integration_time

Reopen reapplies integration time

test_recover_by_reopen_when_transient

Recover by reopen when transient

test_recover_by_usb_restart_after_hang

Recover by usb restart after hang

test_ensure_alive_reopens_when_healthy

Start-of-run reconnect on a working device: just reopen, no pnputil, IT preserved

test_ensure_alive_escalates_to_recover_when_reopen_fails

The device is hung: reopen cannot enumerate it, so ensure_alive falls through to recover()

test_recover_gives_up_when_device_gone

Recover gives up when device gone

test_recover_without_usb_restart

Recover without usb restart

test_instance_id_parsing_filters_by_vid

Instance id parsing filters by vid

Motor-state safety

test_stagestate.py · 7 tests

StateTests

test_roundtrip_and_no_record

Roundtrip and no record

test_power_cycle_signature_is_reported

What the bus looked like on 2026-08-26 after the USB replug: polariser homed to 0,

test_small_settling_is_not_an_anomaly

Small settling is not an anomaly

test_wraparound_compare

Wraparound compare

test_status_only_without_ppd

Status only without ppd

test_protect_and_velocities

Protect and velocities

test_corrupt_record_is_ignored

Corrupt record is ignored

Analysis and reflectance

test_analysis.py · 9 tests

FresnelTests

test_normal_incidence_and_brewster

Normal incidence and brewster

test_slab_inversion_roundtrip

Slab inversion roundtrip

test_bk7

Bk7

SiliconTests

test_tables

Tables

ReflectanceTests

test_recovers_sample_reflectance_with_db_and_different_it

Recovers sample reflectance with db and different it

test_saturated_pixels_are_masked

Saturated pixels are masked

test_missing_dark_and_missing_db

Missing dark and missing db

MaskRegressionTests

test_invalid_wavelengths_exported_as_nan

Si tabulation is invalid below 380 nm: save_csv / at_wavelength must emit NaN there,

test_zero_db_denominator_is_masked_not_inf

A zero double-beam denominator must yield NaN (masked), never inf leaking into R

Unattended tools (monitor / cycle / restore)

test_monitor.py · 11 tests

MonitorTests

test_clean_run_records_state

Clean run records state

test_read_failure_is_recovered

Read failure is recovered

test_read_failure_without_recovery_stops_cleanly

Read failure without recovery stops cleanly

test_state_anomaly_blocks_motion

The fake bus starts at sample 185 / arm 44 -> consistent

RestoreTests

test_report_only_moves_nothing

Report only moves nothing

test_arm_requires_confirmation

Arm requires confirmation

test_safe_and_arm

Safe and arm

CycleTests

test_move_sets_end_at_baseline

Move sets end at baseline

test_dry_run

Dry run

test_read_failure_is_recovered

Read failure is recovered

test_read_failure_without_recovery_aborts_cleanly

Read failure without recovery aborts cleanly

GUI end-to-end (demo hardware)

test_gui_demo.py · 8 tests

GuiDemoTests

test_01_window

Window

test_02_parity

Parity

test_03_lock

Refusals while a sequence is running (no bus needed: the guard fires first)

test_03a_lock_keeps_running_live_read_stoppable

Review regression: during a sequence the Stop live / Stop monitor buttons must stay usable

test_03b_display_path_never_raises

B display path never raises

test_04_tour

Tour

test_05_layout_audit

Layout audit

test_06_render

Render

GUI theme and widgets

test_ui_theme.py · 13 tests

ThemeTests

test_apply_theme

Apply theme

test_px_scaling

Px scaling

test_styles_defined

Accent buttons must not look like default grey Tk buttons

test_card

Card

test_status_pill

Status pill

test_disclosure

Disclosure

test_banner

Banner

test_readout

Readout

test_statusbar

Statusbar

test_form_row

Form row

test_empty_state_and_page_header

Empty state and page header

test_tooltip_and_bind_enter

Tooltip and bind enter

test_mpl_theme

Mpl theme

Demo-hardware harness

test_demo_hw.py · 9 tests

DemoBusTests

test_never_touches_real_hardware

Never touches real hardware

test_queries_and_motion

Queries and motion

test_home_protection

Home protection

test_anomalies

Anomalies

test_raw_query

Raw query

DemoSpecTests

test_open_and_integration_time

Open and integration time

test_frames

Frames

test_failure_and_recovery

Failure and recovery

SeedTests

test_seed_demo_data

Seed demo data